Python Practice: Sets

Part 3 of 5 | 10 questions | Beginner to Expert

RULES: No for/while loops, no comprehensions, no recursion, no map/filter/reduce, and no hardcoded answers. Use indexing, slicing, operators, built-in functions, and data structure methods.

LEVEL: BEGINNER

Q21. Unique Values

```
numbers = [10, 20, 10, 30, 40, 20, 50]
```

Convert to a set and count unique elements.

Q22. Set Operations

```
a = {1, 2, 3, 4, 5}
b = {4, 5, 6, 7, 8}
```

Find union, intersection, a - b, and symmetric difference.

Q23. Add and Remove

```
skills = {"Python", "SQL", "AWS"}
```

Add Snowflake and Databricks; safely remove AWS.

LEVEL: INTERMEDIATE

Q24. Common Students

```
python_students = {"Ali", "Sara", "Ahmed", "Zain"}
sql_students = {"Sara", "Ahmed", "Hamza", "Usman"}
```

Find students taking both and exactly one course.

Q25. Subset and Superset

```
required_skills = {"Python", "SQL"}
candidate_skills = {"Python", "SQL", "AWS", "Snowflake"}
```

Check required skills are a subset and candidate skills a superset.

Q26. Missing Numbers

```
numbers = {1, 2, 4, 6, 8, 10}
```

Find missing integers from 1 through 10 using set operations.

Q27. Three-Set Analysis

```
python = {"Ali", "Sara", "Ahmed", "Zain"}
sql = {"Sara", "Ahmed", "Hamza"}
aws = {"Ahmed", "Zain", "Hamza"}
```

Find students in all three, exactly Python and SQL (not AWS), and at least two.

LEVEL: ADVANCED

Q28. Transaction Analysis

```
day1 = {101, 102, 103, 104}
day2 = {103, 104, 105, 106}
day3 = {104, 106, 107, 108}
```

Find IDs appearing on all days, exactly one day, and at least two days.

Q29. Access Control

```
required_permissions = {"read", "write", "execute"}
user_permissions = {"read", "write", "delete"}
```

Find missing and extra permissions and test full authorization.

Q30. Customer Segmentation

```
website = {"C1", "C2", "C3", "C4", "C5"}
mobile = {"C3", "C4", "C6", "C7"}
store = {"C2", "C4", "C7", "C8"}
```

Find customers on exactly one, exactly two, and all three platforms.